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Cross link interference mitigation for full duplex wireless systems

Abstract

Cross Link Interference (CLI) within a wireless network is mitigated by characterizing a communication path that introduces the CLI on an uplink resource. Mitigation is achieved by receiving, from a neighboring device, information related to a downlink resource scheduling decision. The neighboring device then delays its downlink resource transmission. Transmissions received on an uplink resource may then be corrected.

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Background/Summary

TECHNICAL FIELD

(1) This patent relates to mobile wireless communication systems, and more particularly relates to mitigation of cross link interference in networks that assign uplink and downlink resources in full duplex.

BACKGROUND

(2) The Third Generation Partnership Project (3GPP) Radio Access Network (RAN) working group recently approved its Release 18, generally considered as the initial version of what is now known as the 5G Advanced wireless communication specification. Release 18 is expected to significantly boost 5G performance and support a wide variety of new use cases. These include new operational frequency bands, energy savings, network coverage, mobility improvements, Multiple Input Multiple Output (MIMO) antennas, and positioning services. Support is also provided for Ultra Reliable Low Latency Communication (URLLC), Reduced Capability (RedCap) devices, broadcast services, and Non-Terrestrial Networks (NTNs).

(3) Release 18 continues to treat radio spectrum as a scarce resource. Therefore, further enhancements enable greater flexibility in the use of spectrum. One of these enhancements is the ability for a base station (gNodeB) to operate in full duplex mode. More specifically, Release 18 considers the feasibility of allowing the coexistence of downlink and uplink at the same time within a conventional Time Division Duplex (TDD) channel bandwidth. With one suggested approach, the TDD channel bandwidth is divided into non-overlapping frequency resources, and timeslots are divided between downlink and uplink resources.

SUMMARY OF PREFERRED EMBODIMENTS

(4) Preferred embodiments of the methods, apparatus and systems described herein provide for assignment of uplink and downlink radio resources such that intra-operator and inter-operator interference is mitigated.

(5) More particularly, a wireless network may establish wireless communication between base station and one or more user equipment devices. An uplink resource and a downlink resource are determined, each specified by a respective time domain resource and frequency domain resource. These resources may be chosen to enable full duplex operation at least part of the time.

(6) In some implementations, Cross Link Interference (CLI) imposed by the uplink resource may be cancelled. Cancelling the CLI may involve characterizing the communication path that introduces the CLI (such as by determining its transfer function). The base station is also informed of a downlink scheduling decision by a neighboring base station. The neighboring base station may then delay its downlink transmission, say by at least a single time slot, so that the base station may correct the radio frequency signal it receives on the uplink.

(7) The Transmissions May be Corrected from

$$X1*h1+Y2*h21$$

(8) where * indicates convolution, h1 is the uplink resource estimate, X1 is the signal transmitted on the uplink resource, and Y2 is the downlink signal transmitted by the neighboring basestation, and h21 is the characterization of the communication path that introduces the CLI.

(9) A relative spacing between the frequency resource allocated for the uplink and the frequency resource allocated for the downlink may optionally depend on a Channel Quality Index (CQI) parameter or some other indicator of radio frequency (RF) signal strength. Communication

between the base station and user equipment device using the uplink resource and the downlink resource is then enabled.

(10) If communication between the base station and the user equipment is of the Ultra Reliable Low Latency Communication (URLLC) type, then communications are not delayed. Instead, the uplink frequency resource is chosen from a first group of resources. The downlink frequency resource is assigned from within a second group, but its position within the second group is chosen to be as far from the uplink frequency resource as possible, and regardless of the CQI.

(11) In some implementations, an Artificial Intelligence/Machine Learning (AI/ML) engine may further determine if a particular user equipment device, even though it is experiencing low CQI, is unlikely to be experiencing interference from other devices. When so indicated by the AI/ML engine, the spacing between the downlink and uplink frequency resources may be closer than it would be otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Additional novel features and advantages of the approaches discussed herein are evident from the text that follows and the accompanying drawings, where:

(2) FIG. 1 is a block diagram of a mobile wireless communication system that assigns uplink and downlink resources to mitigate interference.

(3) FIG. 2 illustrates a prior art Third Generation Partnership Project (3GPP) Release 18 duplexing evolution scheme to assign uplink and downlink resources.

(4) FIG. 3 illustrates intra-operator Cross-Link Interference (CLI).

(5) FIG. 4 illustrates a method to reduce the CLI illustrated in FIG. 3.

(6) FIG. 5 shows User Equipment (UE) devices located in different positions relative to a gNodeB.

(7) FIG. 6 is an example assignment of uplink and downlink resources.

(8) FIG. 7 illustrates interference between a gNodeB and a neighboring gNodeB.

(9) FIG. 8 illustrates methods to mitigate interference when an estimate of the neighboring gNodeB channel is available.

(10) FIG. 9 illustrates how uplink and downlink for Ultra Reliable Low Latency Communication (URLLC) traffic may be allocated.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENT(S)

(11) FIG. 1 is a block diagram of a wireless communication system **100** according to one embodiment. The wireless communication system **100** includes multiple user equipment (UE) devices **110-1**, **110-2**, . . . , **110-m**, a base station antenna **112** and a base station **120**. The base station **120** enables the UEs **110** to communicate with other UEs **110** or to send and receive data and/or voice via other networks **190** such as the Internet and/or voice networks. In one embodiment, base station **120** may provide service to UEs **110** located in a particular cell or cell sector.

(12) The base station **120** includes a transceiver **121**, Analog to Digital (ADC) and Digital to Analog (DAC) converters **122**, Digital Signal Processor(s) (DSPs) **123**, memory resources **124**, and processing resources **125**. The transceiver **121** transmits and receives mobile communication signals to and from the UE **110** via antenna(s) **112**, and to and from other base stations, and to and from other communication systems **190**. The ADC and DAC **122** convert the analog signals needed by the transceiver **121** to digital signals processed by the DSPs **123**.

(13) The memory resources **124** include one or more computer readable media that store software instructions for establishing a mobile communication network with the base station **120**. The processing resources **125** execute the instructions stored in one or more computer readable media of the memory resources **124** to process the signals provided by and to the DSPs **123**. As set forth

in more detail below, execution of the software instructions also causes the Radio Resource Management **140** and Interference Mitigation **150** to establish and assign uplink and downlink resources in a particular way. As will be explained in more detail below, Interference Mitigation **150** may include functions such as Channel Quality Measurement **152**, Threshold Settings **153**, Channel Estimates **154** and/or Uplink/Downlink (UL/DL) assignment **156**. Artificial Intelligence (AI)/Machine Learning (ML) engine **160** may also assist with Interference Mitigation **150** and be implemented by software instructions executing on the same or different processing resources **125** and memory resources **124**.

(14) The memory resources **124** and processing resources **125** may be implemented as one or more special-purpose or general-purpose processors. Such special-purpose processors may include processors that are specifically designed to perform the functions detailed herein. Such special-purpose processors may be Application Specific Integrated Circuits (ASIC)s or Field Programmable Gate Arrays (FPGA)s which are general-purpose components that are physically and electrically configured to perform the functions detailed herein. Such general-purpose processors may execute special-purpose software that is stored using one or more memory resources **124** such as non-transitory processor-readable mediums, including random access memory (RAM), flash memory, a hard disk drive (HDD), or a solid state drive (SSD).

(15) Base station **120** may include a 5G New Radio (NR) compliant gNodeB and use a 5G compliant Radio Access Technology (RAT) and Core Mobile Network **170** infrastructure. Core Mobile Network **170** may be in communication with other neighboring gNodeBs **180-1**, **180-2** . . . **180-g**.

(16) However other RATs and other types of network infrastructure can be used. For example, if system **100** is a 4G Long Term Evolution (LTE) network, the base station **120** may be an eNodeB. Therefore, while mobile communication system **100** in one embodiment implements 5G and gNodeBs, the uplink and downlink on bandwidth part schemes detailed herein are applicable to other types of wireless systems **100**, such as a 4G Long Term Evolution (LTE) wireless networks, that uses eNodeBs in place of gNodeBs. Also, system **100** in the described embodiments operates according to the 5G New Radio (NR) radio access technology (RAT). However, in other embodiments, a different RAT may be used, such as 4G Long Term Evolution (LTE), Third Generation (3G) or some other RAT.

(17) An example use case for some embodiments is where the wireless communication system **100** is owned and/or operated by a Mobile Network Operator (MNO) or other wireless system operator. The operator of system **100** (which may also be referred to as an “enterprise”) typically directly owns or controls all the elements necessary to sell and deliver wireless services to an end user of UEs **110**, including radio spectrum license(s), operation of wireless network infrastructure components, back haul, provisioning systems, billing, and customer care

(18) The operator may also offer access to the spectrum by other enterprises such as Mobile Virtual Network Operators (MVNOs) who provide service to the public, Non-Public Network (NPN) providers, or private organizations such as a corporation, a municipality, a university campus, etc.

(19) UEs **110** are various types of wireless computerized devices. For instance, UEs **110** can be smart phones, cellular phones, laptop computers, tablet computers, gaming devices, smart home devices, Internet of Things (IoT) devices, or any other computerized device configured to use the appropriate RAT to communicate with base station **120**.

(20) UEs **110** may also include one or more access points (APs) that further provide network access to one or more other devices. For instance, some devices may be able to communicate wirelessly with an AP via Wi-Fi or Bluetooth or Near Field Communication (NFC). The AP may communicate locally with devices using Wi-Fi and communicate with base station **120** using a different RAT.

(21) As described in more detail below, functions including BWP Management **140** and Interference Mitigation **150** may be provided directly by gNodeB **120**, or by Core Mobile Network

170, or some combination of the two, or by a separate function in communication with Core Mobile Network **170** or gNodeB **120**.

(22) More particularly, Radio Resource Management **140** and Interference Mitigation **150** are responsible for scheduling communications between the base station **120** and UEs **110**. According to 5G Release 18, Physical Resource Blocks (PRBs) may be allocated across the radio bandwidth assigned to a base station **120**. Each PRB may define a time domain resource, such one or more timeslots, on a particular frequency resource (such as one or more subcarrier frequencies). PRBs may be assigned as uplink resources and downlink resources. The number of PRBs scheduled for a given base station **120** may depend on the overall bandwidth available to the operator of the system **100** and on the subcarrier spacing (SCS) within that bandwidth.

(23) As mentioned previously, duplex operation can be provided by allowing the coexistence of downlink and uplink resources within the bandwidth of a given Time Division Duplex (TDD) channel. With one approach, the available TDD channel bandwidth is divided into non-overlapping frequency resources, and then timeslots are further assigned as to uplink and downlink functions. In an embodiment herein, the assignment of PRBs to either uplink or downlink may be implemented by the UL/DL assignment **156** function as part of Interference Mitigation **150**.

(24) FIG. 2 shows one example prior art scheme **200** for allocating uplink and downlink PRBs to provide subband full duplex (SBFD) within a carrier or in multiple aggregated carriers. This particular scheme supports the coexistence of multiple downlink and uplink resources within a TDD channel **220**.

(25) PRBs are assigned such that the TDD channel **220** is divided into multiple nonoverlapping PRBs, with some PRBs designated for uplink and others designated for downlink. In many use cases, such as Internet access, streaming media applications and the like, downlink traffic from the base station **120** to the UEs **110** is greater than uplink traffic from the UEs **110** to the base station **120**. Therefore it may be advantageous to assign more downlink channels than uplink channels in each timeslot.

(26) In this example implementation, the bandwidth for TDD channel **220** may be available within a single radio frequency carrier or may comprise multiple aggregated carriers. The TDD channel bandwidth, in one example, may be 20 MHz which is then divided into 106 PRBs, each 180 kHz wide. During first timeslot t1, a first subgroup **210-D-1** of these 106 PRBs is assigned to downlink, a second group of frequency resources **210-U-1** is assigned to uplink, and a third group of these frequency resources **210-D-2** is assigned to downlink. In this example, the first group **210-D-1** may include forty (40) PRBs, the second group **210-U** may include thirty six (36) PRBs, and the third group **210-D-2** may include the remaining thirty (30) PRBs.

(27) The assignments in timeslots t2, t3, and t4 could be the same as the assignments in timeslot t1. Timeslot t5 is dedicated to uplink only in this scenario.

(28) With this approach, uplink PRBs in timeslots t1 to t4 are deployed using only the frequency resources located in the center block of the TDD channel, with downlink resources deployed using the frequency resources on either side of the center block.

(29) The pattern then could dynamically change in some defined way on each timeslot, or repeat on a semi-persistent fashion for subsequent timeslots after t5.

(30) This approach therefore permits implementing Frequency Division Duplex (FDD) during timeslots t1-t4 (by assigning uplink and downlink resources simultaneously using different frequency resources in a given timeslot). This is generally the preferred method for communication with UEs located in most areas of a typical cell.

(31) This approach for assignment of uplink and downlink resources also permits implementing Time Division Duplex (TDD), such as by assigning uplink and downlink resources on the same frequency resources but in different timeslots.

(32) FDD can be the preferred approach for UEs that are located on the outer edge of a cell. This is because cell edge devices often operate with limited link budget and at relatively high power, and

are more susceptible to interference from other base stations. Cell edge devices are thus more reliant on schemes for improving reception such as by resolving multipath. By implementing both uplink and downlink on the same subcarrier for these cell edge UEs, channel propagation estimates measured on a downlink frequency resource will more accurately reflect necessary adjustments to the transmit parameters for uplink transmissions which occur using the same frequency resource.

(33) However, TDD provides a number of advantages compared to FDD in some instances. TDD generally provides more flexible and efficient use of the available download and upload resources based on traffic volumes. With TDD, the assignment of download and upload frequency resources can more easily be made semi-static, or even dynamic based on measured demand. In addition, many of the newly available spectrum bands for 5G are located at higher carrier frequencies and provide significantly higher bandwidth as compared to the legacy FDD frequency bands. Furthermore, MIMO performance is improved with TDD. Because the same subcarrier is used for both uplink and downlink, any adjustments derived from receiver signal processing are more accurately applied to the transmit side.

(34) TDD has a disadvantage when it comes to Ultra Reliable Low Latency Communication (URLLC). Because TDD exhibits a slight time delay (due to the fact that transmit slots are only available part of the time) compared to FDD (where transmit slots are always available), some TDD packets will be slightly delayed.

(35) TDD also exhibits reduced link budget. In particular, lower signal-to-noise ratio is available in TDD than with FDD when transmitting at peak power. That is because a transmitter operates over one-half the available bandwidth all of the time in FDD, but transmits over the entire bandwidth half of the time in TDD. As a result, the transmit power amplifier is only active half of the time, effectively reducing average radiated power in half.

(36) Furthermore, TDD suffers from inter-cell interference more readily than FDD. This is due to the fact that a downlink at the cell edge can be impacted by out of band emissions of the uplink of adjacent cells or sectors operating on neighboring frequency resources.

(37) Returning attention briefly to FIG. 1, the system **100** may include a Resource Management **140** function. Resource Management **140** may be implemented for a 5G NR system **100** using the processing and memory resources within gNodeB **120** and/or by processing and memory resources separate from and in communication with gNodeB **120**. For other types of networks, Resource Management **140** can also be part of network management (NM) or system-level Radio Resource Management (RRM) functions.

(38) Resource Management **140** configures the gNodeB **120** to divide available TDD channel bandwidth into PRBs, with each PRB specified by a time domain resource and a frequency domain resource. The PRBs are then used by the base station **120** concurrently for communication with different pieces of user equipment (UE) **110**. The frequency domain resource for each individual PRBs may be allocated from a single subcarrier frequency or multiple aggregated subcarriers, and may be different from the frequency resources assigned to other PRBs.

(39) As explained in more detail below, the available PRBs are assigned as either uplink or downlink PRBs by UL/DL resource management **156**, in a particular way, as part of interference mitigation **150**.

(40) FIG. 3 illustrates a typical situation where interference is possible. Here a pair of base stations gNodeB1 **120-1** and gNodeB2 **120-2** are servicing multiple UEs **110** using their assigned PRBs. UE1 (**110-1**) and UE2 (**110-2**) are relatively close to one another, but serviced by different gNodeBs. UE1 is located near the edge of the area serviced by gNodeB 1 (**120-1**) and UE2 is near the edge of the area serviced by another neighboring gNodeB 2 (**120-2**). gNodeB 1 and gNodeB 2 may be servicing different cell sites or different sectors of the same site.

(41) The frequency spectrum plot **310** shown next to UE1 and UE2 shows relative signal power for the uplink (UL) and downlink (DL) PRBs near UE1 and UE2 in a particular timeslot. For example, the timeslot may correspond to timeslot t1 in FIG. 2 where PRBs in the middle of the frequency

band are assigned to uplink and PRBs on one or both sides thereof are assigned to downlink.

(42) At this instant in time, UE1 is assigned an uplink PRB, transmitting to gNodeB 1, and UE2 is receiving from gNodeB 2 on two assigned downlink PRBs.

(43) Note that a possible Cross Link Interference (CLI) situation exists at UE2. As spectrum plot **310** shows, the received RF signal strength on the uplink PRB radiated by UE1 may be greater in power than the downlink PRB signal received by UE2. Thus UE2 is a possible “victim” and UE1 a possible “aggressor”.

(44) Cross Link Interference is also possible at gNodeB 1. As shown in spectrum plot **312**, gNodeB 2 radiates on the downlink PRB with relatively high power so that it may reach UE2. However, such a strong signal may also reach gNodeB 1, and thus the received RF signal strength on the downlink PRB(s) (as radiated by gNodeB 2) may be greater than the signal strength received on the uplink from UE1. Here UE1 is considered a possible “aggressor” and gNodeB 1 a possible “victim”.

(45) Consider that a similar CLI issue may occur in the same time slot for another UE3 **110-3** also serviced by gNodeB 1 on uplink **326**, even though it is located away from UE2. Similarly, gNodeB 2 may experience Cross Link Interference on its own uplink, such as for yet another UE4 **110-4**.

(46) To summarize, it is possible that uplink PRBs transmitted by some UEs may interfere with downlink PRBs assigned to other nearby UEs, and downlink PRBs transmitted by a gNodeB may interfere with downlink PRBs assigned by other gNodeBs. This interference may become more pronounced as UEs move towards the edge of the cell (or sector).

(47) As the frequency spectrum plots **310**, **314** show, the level of interference tends to decrease as the separation between uplink and downlink PRB channels increases. A possible solution is for UL/DL assignment **156** to allocate PRBs such that relatively weak downlink PRB's (such as for UEs located at a cell edge) are scheduled to use frequency resources further away from the uplink PRBs in the same time slot. One may use a parameter, such as the value of a received signal strength indicated by a Channel Quality Indicator (CQI), to determine how to assign PRBs to subchannels. CQI may be determined by CQI Measurement function **152** as part of Interference Mitigation **150** (FIG. 1).

(48) FIG. 4 illustrates a more general example of cell/sector where gNodeB **120** is servicing a number of UEs. At first UE1 **110-1** and second UE 2 **110-2** are located near the cell (or sector) edge in region **404** and adjacent one another. A third UE3 **110-3** is also located near a cell edge in region **404**, but away from UE1 **110-1** and UE2 **110-2**. UE5 **110-5** is located relatively close to the gNodeB **120** in region **408**. UE 6 **110-6** is located somewhere in region **406** between UE4 and UE 3 such that it is neither close to gNodeB nor is it at the cell/sector edge.

(49) In this particular timeslot, UE1 and UE6 are operating in the uplink direction, and UE 2, UE3, UE4 and UE5 are operating in the downlink direction. It should be understood that FIG. 4 shows a single timeslot and that uplink and downlink assignments are potentially different for other timeslots.

(50) The area serviced by this example cell/sector may also be divided into three regions or zones based on distance from the gNodeB, including a low CQI region **404** (furthest from the gNodeB), high CQI zone **408** (closest to the gNodeB) and a middle CQI region (between the two other region). In other example embodiments, more than three such CQI regions or fewer than three CQI regions may be defined.

(51) FIG. 5 is one possible approach for Interference Mitigation **150** to assign uplink PRBs and downlink PRBs to the UEs shown in FIG. 4. This approach arranges the PRBs that utilize frequency resources (such as subcarriers) located between a lower limit **514-1** and an upper limit **514-2**. The PRBs are further arranged into three defined groups. A first group of PRBs **502** are assigned for uplink, and the remaining PRBs **503-1**, **503-2** on either side of the first group **502** are assigned for downlink. Each group **502**, **503-1**, **503-2** of PRBs may include two or more PRBs that utilize frequency domain resources such as subcarriers that are contiguous or at least neighboring in

frequency.

(52) The extent of the first group **502** assigned for uplink ranges between a subcarrier **520-1** and subcarrier **520-2**. Downlink group **503-1** includes frequency resources between a lower subcarrier limit or threshold **514-1** and subcarrier **520-1**. Downlink group **503-2** includes frequency resources between subcarrier threshold **520-2** and upper subcarrier threshold **514-2**. These thresholds may be determined by Threshold Settings **153**.

(53) Note also that downlink PRBs **503-1** may be further split into different zones. In this example, there are three downlink zones defined by thresholds **516-1** and **516-2**, to provide a downlink zone **524-L-1** located between threshold **514-1** and threshold **516-1**, downlink zone **524-M-1** located between thresholds **516-1** and **518-1**, and downlink zone **524-H-1** located between thresholds **518-1** and **520-1**.

(54) In general, Interference Mitigation **150** prioritizes downlink zones **524-L-1** and **524-2** for use by low CQI UEs located furthest away from uplink zone **502**, downlink zones **524-H-1** and **524-H-2** for use by high CQI UEs closest to uplink zone **503**, and middle CQI zones **524-M-1** and **524-M-2** for other UEs.

(55) However, if there are an insufficient number of low CQI UEs to fill the downlink zones **524-L-1** and **524-L-2**, the downlink resources for the middle CQI UEs can be scheduled there, and then if they are still not filled, the high CQI UEs can be scheduled there.

(56) In other words, the scheduler may attempt to first schedule downlink PRBs for low CQI UEs in the zones **524-1**, **524-2** located furthest away from the uplink zone **502**, and then schedule downlink PRBs for the middle CQI UEs in both the low zones **524-L-1**, **524-L-2** and the middle downlink zones **524-M-1**, **524-M-2**. The downlink resources for high CQI UEs may then be scheduled in any of the remaining downlink zones.

(57) Similarly, downlink PRBs **503-2** may be further split into three downlink zones **524-H-2**, **524-M-2**, **524-L-2**.

(58) UE1 and UE6 operate in this time slot in the uplink direction. Therefore they have PRBs assigned them from the center zone **502**.

(59) Interference mitigation function **150**, having information from CQI Measurement **152** as to the relative receive signal strengths for the various UEs may assign downlink PRBs as follows.

(60) UE2 has reported a low CQI (e.g., it is located far from the gNodeB in region **404** of FIG. 4) and thus is assigned a downlink PRB in zone **524-L-1**.

(61) Similarly, since UE3 also reports a low CQI, it too is assigned a downlink PRB in zone **524-L-2**.

(62) UE5 is detected as having high CQI (because it is located in region **408**) and thus has been assigned a downlink PRB in zone **524-H-1** closest to uplink zone **502**.

(63) UE4, having a CQI that is neither high nor low (being located in middle region **406**) is assigned a downlink PRB in zone **524-M-2**.

(64) In some implementations, there may be a guard band **530** maintained between low zone **524-L-2** and mid zone **524-M-2** where no PRBs are actively allocated. This may assist with ensuring the receive signal strength at devices such as UE4 is sufficient even while devices such as UE3 are active.

(65) Note also that the uplink for UE6, having the larger CQI, is less susceptible to interference from downlink PRBs than, say, UE1. Therefore UE6 may be assigned an uplink PRB located closer to the edge of the center zone **502**.

(66) FIG. 6 illustrates a method **600** which may be carried out by interference mitigation **150** of FIG. 1 schedule downlink PRBs based on CQI or some other indicia of relative signal strength.

(67) In a first state **602** the size of the uplink zone **502** is determined. This may be based on how much uplink traffic is expected versus downlink traffic.

(68) In state **604** the gNodeB may then determine the location of the different downlink PRB thresholds **516-1**, **518-1**, **516-2**, **518-2**. Threshold Setting **153** may fix the location of these

threshold zones in advance, or may adapt to the present traffic observed by the gNodeB. For example, when there are more UEs expected to have relatively low CQI, then thresholds **516-1**, **516-2** may be adjusted inward to make more downlink PRBs available for the low CQI UEs—in other words making the low CQI zone(s) **504-L-1**, **504-L-2** larger. Also, when most of the UEs assigned to the gNodeB are expected (or measured) to have high CQI, then thresholds **518-1**, **518-2** might be shifted outward to allow greater space for handling UEs with high CQI.

(69) In a next state **606**, data is collected by CQI Measurement **152** for each active UE. The CQI may be measured by each UE during a downlink PRB and reported back to the gNodeB. For 3GPP compliant networks, the CQI report may be via a control channel message such as a Dedicated Control Channel (DCCH) message. In other scenarios, CQI may be reported differently, or measured by the gNodeB directly from the RF signal it receives on an uplink channel associated with a given UE.

(70) It should be understood that although not shown as a separate state in FIG. **6**, the CQI measurements may be fed back to Threshold Setting **153** as part of state **604**, to enable adaptively setting the downlink thresholds **516-1**, **518-1**, **516-2**, **518-2** (see FIG. **5**) based on current conditions.

(71) In state **608**, downlink PRB's for UEs that are further away from the gNodeB (having weaker downlink power) may be scheduled further away from the uplink PRBs.

(72) In state **610**, UEs that are located closer to the gNodeB may have their downlink PRB's located somewhat closer to the uplink PRBs than for low CQI UEs. It may, in some instances, be desirable to include a guard band between the uplink and downlink PRBs, even for these middle range UEs.

(73) In state **612**, UEs that are located closest to the gNodeB and having the strongest CQI have downlink PRBs assigned to be adjacent to the uplink PRBs.

(74) As can now be appreciated, creating isolation between uplink and downlink PRBs may mitigate interference especially for low CQI links.

(75) Other types of interference can also be mitigated using similar techniques for scheduling uplink and download link PRBs taking into account the potential for Cross Link Interference (CLI).

(76) One such scenario is shown in FIG. **7**. In this embodiment, similar to the scenario in FIG. **3**, gNodeB 1 is receiving information on a uplink PRB **720** from UE 1. At the same time, gNodeB 2 is transmitting on downlink PRB **722** to UE2. Note that the radiated power level of downlink transmission by gNodeB 2 does not depend on the distance between gNodeB 2 and UE 2—in other words, the radiated power may be relatively strong even when UE2 is close to gNodeB 2. This in turn may cause gNodeB 1 to also receive the downlink transmission from gNodeB 2, causing Cross Link Interference.

(77) Information regarding the state of the resulting “crosslink” or “out of band” channel **702** between gNodeB 1 and gNodeB 2 can be used to mitigate such CLI. Generally, speaking, if gNodeB 1 knows how to characterize channel **702** (such as channel **702**'s transfer function), and also knows the signal transmitted by gNodeB 2 on downlink **722**, gNodeB 1 can estimate such interference and remove it from the uplink signal **720** received from UE1. This estimate may be provided by Channel Estimate **154** function as part of Interference Mitigation **150**.

(78) FIG. **8** illustrates an example method **800** to implement this technique. In state **801**, gNodeB 1 may determine whether current traffic is identified as Ultra Reliable Low Latency Communication (URLLC). If not, it may proceed to state **802**. However, if URLLC services necessary, then state **810** is entered.

(79) In state **802**, gNodeB 1 receives information that characterizes the cross link channel **702** between itself and gNodeB 2. This may be done by having gNodeB 2 periodically and briefly transmit a known signal (such as on a control channel without other signals) so that gNodeB 1 can periodically detect and characterize channel **702** such as by its determining its transfer function (h21).

(80) In state **804**, gNodeB 2 informs gNodeB 1 of its downlink schedule decision(s) at least one slot ahead of actual downlink transmission(s) **722**.

(81) In state **805**, gNodeB 2 transmits on downlink PRB **722** (delayed by one time slot)

(82) Next, in state **806** gNodeB 1 receives the uplink **720** from UE1. The signal received by gNodeB 1 can be estimated as $X1*h1+Y2*h21$

(83) where * indicates convolution, h1 is the uplink channel, X1 is the signal transmitted by UE1 to gNodeB 1 on uplink PRB **720**, and Y2 is the downlink signal transmitted by gNodeB 2 to UE2 at the same timeslot that UE1 transmits signal X1 to gNodeB1.

(84) As mentioned above, the assumption is that gNodeB2 informs gNodeB1 of its scheduling decision and content of Y2 through link **702** at least one time slot ahead of time. This information includes the actual complex signal it intends to send (including the payload information, coded bits, etc) as well as the PRB(s) in which Y2 is scheduled to be transmitted.

(85) In state **808**, given that gNodeB 1 knows h21 and Y2 (the signal transmitted by gNodeB 2) it can determine $Y2*H21$ and thus cancel the CLI (e.g., the “out of band transmission”) introduced into Y2 by gNodeB 2.

(86) Alternatively, for the URLLC traffic the one slot delay imposed on gNodeB 2 in state **806** may not be acceptable.

(87) In state **812** such URLLC traffic may not be delayed, but instead have its downlink PRBs scheduled in the low CQI zone(s), far from the uplink zones to increase guard band between them and thus mitigate interference.

(88) An example of such downlink PRB allocation for URLLC is shown in FIG. **9**. Here the downlink PRBs **912-1**, **912-2** are located in zones **924-1**, **924-2** away from the uplink zone **902**—regardless of the measured CQI for UE1. Non-URLLC downlinks may be assigned in the other zones **925-1**, **925-2**. Thus, even when the CQI for URLLC may be relatively high, the downlink PRBs appear at the outer edges of the BWP **900** to minimize CLI from the uplink PRB(s) **902**.

(89) In some implementations, an Artificial Intelligence (AI) and/or Machine Learning (ML) based processing resource can implement aspects of interference mitigation. For example, it may detect that a UE located on the cell edge and thus operating with low CQI is not actually interfering with other UEs. In that case, the low CQI UE may be scheduled anywhere.

(90) Referring back to FIG. **1**, one example implementation uses an AI/ML based engine **160** to analyze data it receives from CQI measurements **152** and/or channel estimates **154** to detect whether a particular device (UE) is not impacted from interference created by uplink transmissions of the devices. In other words, the AI/ML engine may determine that the particular UE device is located at a cell edge, and that there is no other UE device currently transmitting on the uplink close to this particular device. Example situations may be as were depicted in FIG. **4** and/or FIG. **7**. Although UE3 of FIG. **4** is located near the cell edge, it is not experiencing interference from any of the other active UEs in the cell. Similarly, U3 in FIG. **7** is not experiencing any interference from gNodeB2. The AI/ML engine can thus detect these situations, and inform UL/DL assignment that UE3's downlink PRBs can be scheduled within any of the available downlink zones (low CQI zones **524-L-1**, **524-L-2**, mid CQI zones **524-M-1**, **524-M-2**, or high CQI zones **524-H-1**, **524-H-2**.

(91) AI/ML engine **160** may include methods to train a machine learning model to determine when an UE is located at the edge of a cell but is not likely to be experiencing interference from other UEs. A set of training data can include characteristics on a large number of CQI measurements and/or channel estimates for cell edge UEs, along with data about where the other UEs are located when those measurements were taken.

(92) For example, one possible machine learning implementation can involve a neural network being trained based on some or all of these characteristics. These characteristics can serve as input features to the neural network. The neural network may then perform a classification based on the features to inform UL/DL Assignment **156** that is it free to assign the particular UE a PRB outside

of the usual low CQI zone.

(93) In some embodiments, other forms of machine learning may be used to estimate whether the cell edge UE is likely to be experiencing interference from other UEs, and then inform the UL/DL Assignment 156 of that determination. UL/DL Assignment may then use that determination of the likelihood of interference to choose the appropriate PRB zone for the UE.

(94) The methods, systems, and devices discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims.

(95) Specific details are given in the description to provide a thorough understanding of example configurations (including implementations). However, configurations may be practiced without these specific details. For example, well-known circuits, processes, algorithms, structures, and techniques have been shown without unnecessary detail in order to avoid obscuring the configurations. This description provides example configurations only, and does not limit the scope, applicability, or configurations of the claims. Rather, the preceding description of the configurations will provide those skilled in the art with an enabling description for implementing described techniques. Various changes may be made in the function and arrangement of elements without departing from the spirit or scope of the disclosure.

(96) Also, configurations may be described as a process which is depicted as a flow diagram or block diagram. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional states or steps not included in the figure.

Furthermore, examples of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks may be stored in a non-transitory computer-readable medium such as a storage medium. Processors may execute the program code to perform the described tasks.

(97) Having described several example configurations, various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the disclosure. For example, the above elements may be components of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after the above elements are considered.

(98) The above description has particularly shown and described example embodiments. However, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the legal scope of this patent as encompassed by the appended claims.

Claims

1. A method for wireless communication, comprising: establishing, by a wireless network, time domain resources and frequency domain resources to implement wireless communication by a base station; allocating an uplink resource and a downlink resource for communication between the base station and a user equipment device, the uplink resource and downlink resource each specified by a respective time domain resource and respective frequency domain resource, the frequency domain resource allocated to the uplink resource and the frequency domain resource allocated to the downlink resource allocated to enable full duplex communication; performing, by the base station,

communication with the user equipment device using the uplink resource and the downlink resource; cancelling Cross Link Interference (CLI) on the uplink resource caused by a neighboring base station by further: characterizing a communication path that introduces the CLI on the uplink resource; informing, by the neighboring base station, of a downlink resource scheduling decision; delaying, by the neighboring base station, a downlink resource transmission; receiving, by the base station, the delayed downlink resource transmission; and correcting transmissions received on the uplink resource based on the characterizing the communication path that introduces the CLI on the uplink resource and the delayed downlink resource transmission.

2. The method of claim 1 wherein characterizing the communication path further comprises: periodically receiving a known signal from the neighboring base station.

3. The method of claim 1 wherein the informing step is performed at a predetermined time before the downlink resource transmission by the neighboring base station.

4. The method of claim 2 wherein the step of correcting transmissions received via the uplink resource further comprises: estimating the transmissions received via the uplink resource as $X1*h1+Y2*h21$ where * indicates convolution, $h1$ is an uplink resource channel estimate, $X1$ is the signal transmitted on the uplink resource, $Y2$ is a known signal transmitted by the neighboring base station, and $h21$ is the characterization of the communication path that introduces the CLI on the uplink resource.

5. The method of claim 1 additionally wherein the frequency domain resources are allocated in groups, with a first group of frequency domain resources having frequencies located in a middle portion of the frequency domain resources, and a second group of frequency resources having frequencies either lower than or greater than the frequency resources in the middle portion.

6. The method of claim 5 additionally comprising: when the communication is identified as Ultra Reliable Low Latency Communication (URLLC), then not performing the step of cancelling CLI on the uplink resource, and instead allocating the frequency domain resource for the uplink resource in the first group of frequency resources, and allocating the frequency domain resource for the downlink resource in the second group of frequency domain resources.

7. The method of claim 1 additionally comprising: determining a Channel Quality Index (CQI) indicative of a radio frequency (RF) signal strength associated with communication between the base station and a user equipment device; wherein a relative spacing between the frequency domain resource allocated for the uplink and the frequency domain resource allocated for the downlink depends on the CQI.

8. The method of claim 7 additionally comprising: determining a second Channel Quality Index (CQI) that depends on a radio frequency (RF) signal strength associated with communication between the base station and a second user equipment; allocating a second uplink resource and a second downlink resource for use by the second user equipment, the second uplink resource and second downlink resource each specified by a respective frequency domain resource and a respective time domain resource, the resources allocated to enable full duplex communication by the second user equipment; wherein the frequency domain resource allocated for the second uplink resource is located in a predetermined portion of the frequency domain resources; wherein a relative spacing between the frequency domain resource allocated for the second uplink resource and the frequency domain resource allocated for the second downlink resource depends on the second CQI; and performing, by a base station, communication with the second user equipment using the second uplink resource and the second downlink resource.

9. The method of claim 8 additionally wherein: when the CQI is below a low threshold, the frequency domain resource selected for the second downlink resource is a frequency domain resource located within a second group of frequency domain resources and located away from the frequency domain resource assigned to the second uplink resource; and when the CQI is above a high threshold, the frequency domain resource selected for the second downlink resource is located within the second group of frequency domain resources and located near the frequency domain

resource assigned to the second uplink resource.

10. The method of claim 1 additionally comprising: performing a machine learning process to analyze one or more characteristics of the communication between the base station and the user equipment device, wherein the one or more characteristics of communication include a likelihood the user equipment device is experiencing interference from neighboring user equipment devices; and further allocating the uplink resource and downlink resource depending on such likelihood.

11. A base station apparatus comprising: a transceiver configured to transmit and receive signals to and from a user equipment device; one or more processors coupled to the transceiver; and one or more computer readable media including instructions that, when executed by the one or more processors, cause the one or more processors to perform a process for: establishing time domain resources and frequency domain resources to implement wireless communication; allocating an uplink resource and a downlink resource for communication between the base station and the user equipment device, the uplink resource and downlink resource each specified by a respective time domain resource and frequency domain resource, the frequency domain resource allocated to the uplink resource and the frequency domain resource allocated to the downlink resource also allocated to enable full duplex communication; performing, by the base station, communication with the user equipment device using the uplink resource and the downlink resource; and cancelling Cross Link Interference (CLI) on the uplink resource caused by a neighboring base station by further: characterizing a communication path that introduces the CLI on the uplink resource; being informed, from the neighboring base station, of a downlink resource scheduling decision; receiving, by the base station, a delayed downlink resource transmission; and correcting transmissions received on the uplink resource based on the delayed downlink resource transmission and the characterizing the communication path that introduces the CLI on the uplink resource.

12. The apparatus of claim 11 wherein the one or more processors further perform the process of characterizing the communication by: periodically receiving a known signal from the neighboring base station.

13. The apparatus of claim 11 wherein the one or more processors are further for informing of the downlink resource scheduling decision at a predetermined time before the downlink resource transmission by the neighboring base station.

14. The apparatus of claim 11 wherein the one or more processors further perform the process of correcting transmissions received via the uplink resource by: estimating the transmissions received via the uplink resource as

$X1 * h1 + Y2 * h21$ where * indicates convolution, $h1$ is an uplink resource estimate, $X1$ is the signal transmitted on the uplink resource, $Y2$ is a known signal transmitted by the neighboring base station, and $h21$ is the characterization of the communication path that introduces the CLI on the uplink resource.

15. The apparatus of claim 11 wherein the one or more processors further perform the process of allocating the frequency domain resources in groups, with a first group of frequency domain resources having carrier frequencies located in a middle portion of the frequency domain resources, and a second group of frequency resources having carrier frequencies either lower than or greater than the frequency resources in the middle portion.

16. The apparatus of claim 15 wherein the one or more processor further perform the process of: when the communication is identified as Ultra Reliable Low Latency Communication (URLLC), not cancelling the CLI on the uplink resource, and instead allocating the frequency domain resource for the uplink resource in the first group of frequency domain resources, and allocating the frequency domain resource for the downlink resource in the second group of frequency domain resources.

17. The apparatus of claim 11 wherein the one or more processors further perform the process of: determining a Channel Quality Index (CQI) indicative of a radio frequency (RF) signal strength associated with communication between the base station and a user equipment device; and wherein

a relative spacing between the frequency resource allocated for the uplink and the frequency resource allocated for the downlink depends on the CQI.

18. The apparatus of claim 17 wherein the one or more processors further perform the process of: determining a second Channel Quality Index (CQI) that depends on a radio frequency (RF) signal strength associated with communication between the base station and a second user equipment; allocating a second uplink resource and a second downlink resource for use by the second user equipment, the second uplink resource and second downlink resource each specified by a respective frequency domain resource and respective time domain resource, to enable full duplex communication by the second user equipment; wherein the frequency domain resource allocated for the second uplink resource is located in a predetermined portion of the frequency domain resources; wherein a relative spacing between the frequency domain resource allocated for the second uplink resource and the frequency domain resource allocated for the second downlink resource depends on the second CQI; and performing, by a base station, communication with the second user equipment using the second uplink resource and the second downlink resource.

19. The apparatus of claim 18 wherein the one or more processors further perform the process of: when the CQI is below a low threshold, the frequency domain resource selected for the second downlink resource is a frequency domain resource located within a second group of frequency domain resources and located away from the frequency domain resource assigned to the second uplink resource; and when the CQI is above a high threshold, the frequency domain resource selected for the second downlink resource is located within the second group of frequency domain resources and located near the frequency domain resource assigned to the second uplink resource.

20. The apparatus of claim 18 additionally wherein the one or more processors for further perform the process of: performing a machine learning process to analyze one or more characteristics of the communication between the base station and the user equipment device, wherein the one or more characteristics of communication include a likelihood the user equipment device is experiencing interference from neighboring user equipment devices; and allocating the uplink resource and downlink resource depending on such likelihood.
